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Beyond Forecast Accuracy: Feasibility-Constrained Demand Planning

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Abstract

Demand planning teams optimize forecast accuracy; operations teams execute plans. These objectives diverge: a more accurate forecast often requires plan changes that exceed operational absorption capacity, creating coordination burden and execution gridlock. This paper reframes forecast-driven demand planning as feasibility-constrained model selection, evaluating forecast accuracy and execution feasibility as dual objectives.

The core contribution is a forecast-to-plan feasibility gate: candidate forecasts are converted into executable planning signals, measured for operational consequence (inventory exposure, plan stability, execution violations, governance burden), and ranked by feasibility-adjusted planning utility rather than accuracy alone. The gate is model-agnostic, enabling fair comparison of accuracy-first, ensemble, smoothing, greedy, and dynamic programming policies.

Empirically, the study draws on DataCo fulfillment records to calibrate data-informed execution-risk severity levels, alongside Favorita and M5 retail demand data. Across 123 evaluation groups, the accuracy-first deployable strategy differs from the operational-loss-optimal deployable strategy in 91.1% of cases. The accuracy-first planning-execution gap rate is 15.4% of periods, compared with 2.7% for the best deployable operational-loss strategy. Ensemble methods offer low-disruption alternatives; dynamic programming and smoothing serve different governance and execution constraints, and strategy selection varies with planning hierarchy and demand sparsity.

The practical implication is direct: deployment decisions should be made at the forecast-to-plan layer, not the forecasting layer. Forecast accuracy remains important but should be treated as a diagnostic metric, separate from operational planning utility. Organizations can improve plan feasibility while sacrificing only modest accuracy, enabling faster, more confident model rollouts.

Keywords

Demand planning, Forecast-to-plan conversion, Operational feasibility, Planning stability, Execution risk.

1 Introduction

1.1 The Planning-Execution Gap

Demand planning is an operational deployment problem, not only a forecasting problem: forecast-driven decisions influence inventory targets, replenishment orders, supplier coordination, warehouse labor, transportation commitments, and planning-system governance. Recent advances in AI-enabled forecasting and time-series foundation models have made more accurate predictions available (Walter et al. 2025; Culot et al. 2024; Makridakis et al. 2022), but prediction accuracy alone does not guarantee deployment success, because operations execute planning signals, not forecast error metrics. A planner cannot execute WAPE or RMSE; the organization executes an inventory target, a staffing plan, or a replenishment quantity, and when forecast-driven plans change faster than supplier lead times, labor cycles, or system change windows can absorb, the forecast and execution layers become misaligned—a planning-execution gap in which the operation cannot execute the plan despite its accuracy.

The gap manifests in four operational symptoms:

1. **Execution violations**: Plans change faster than downstream systems can absorb (e.g., warehouse rescheduling, supplier coordination, system updates)
2. **Governance burden**: Each model change requires validation, approval, and monitoring cycles, delaying rollout
3. **Supplier disruption**: Frequent plan changes force suppliers to expedite orders or renegotiate, increasing costs
4. **Operational gridlock**: Operations teams spend effort coordinating plan changes rather than executing the plan

These symptoms are invisible in accuracy metrics but visible in execution logs, planner workload, and supplier relationships. The operational reality is that forecast accuracy and plan executability are distinct objectives.

1.2 From Forecasting Optimization to Feasibility-Constrained Selection

Forecasting research traditionally optimizes for accuracy—lower WAPE, lower MAE, better leaderboard scores—which is valuable, but a more accurate forecast that creates execution gridlock is operationally worse than a slightly less accurate forecast that executes smoothly.

The solution is to shift the optimization target from *forecast generation* to *forecast deployment*: ask which deployable planning strategy minimizes operational cost under execution constraints, not which model minimizes WAPE. This paper operationalizes that shift as a model-agnostic forecast-to-plan feasibility gate: it converts each candidate forecast into an executable planning signal, scores its operational consequence on four metrics (inventory exposure, plan stability, execution violations, governance burden), ranks strategies by feasibility-adjusted utility rather than accuracy alone, and validates ranking stability across execution-risk scenarios and planning contexts.

1.3 Empirical Evidence and Scope

The empirical design uses three complementary datasets in distinct operational roles. **Favorita** (main proof of concept) is daily retail demand from 54 stores and 33 product families (1,782 planning units) over 4+ years, testing whether the accuracy-feasibility tradeoff appears in richly-contexted, fine-grained planning. **M5** (hierarchy and intermittency robustness) is daily retail demand from 3,049 items across 10 stores with item-store, department-store, and category-store aggregation, testing whether strategy preferences change with granularity and demand sparsity. **Walmart** (business-context and cadence robustness) is weekly store-department sales with markdown, holiday, and macro signals, testing whether weekly cadence and rich context change the tradeoff. Across these datasets, 91.1% of evaluation groups show that the WAPE-best strategy differs from the operational-loss-optimal strategy—not a data artifact, but structural.

1.4 Objectives

This paper pursues five objectives, each also a research contribution: (1) define a forecast-to-plan feasibility gate separating diagnostic accuracy from operational utility; (2) quantify the gap using execution violation rate and normalized planning loss; (3) compare six strategy families under the same feasibility evaluation across data-informed execution-risk scenarios; (4) validate, across Favorita, M5, and Walmart, that strategy rankings shift with scenario, hierarchy, and context; and (5) translate the findings into a five-step deployment protocol operations teams can apply directly.

The remaining sections follow the same arc as the four symptoms in Section 1.1: Section 3 turns the feasibility gate into a mechanism, Section 5 reports how often and why accuracy-first and deployment-best selection diverge, Section 5.5 converts that evidence into a five-step deployment protocol, and Section 5.6 closes the loop by testing whether the protocol holds under tighter execution risk, coarser and finer planning grains, and a second dataset with a different cadence and business context.

2 Literature Review

2.1 AI-Enabled Demand Planning and Forecasting Progress

Recent supply-chain research shows rapid progress in AI-enabled demand planning. Walter et al. (2025) identify AI tools, supply-chain applications, and digital transformation as major knowledge clusters; Culot et al. (2024) survey empirical AI adoption and emphasize connecting AI advancement to operational outcomes; Vlachos and Reddy (2025) identify decision-support, execution optimization, and control monitoring as high-value areas. Forecasting benchmarks show the same trend: the M5 competition demonstrates modern retail forecasting at scale across hierarchies and intermittency (Makridakis et al. 2022), and recent time-series foundation models suggest forecasting pipelines are moving toward broader zero-shot and cross-domain capability (Das et al. 2024; Ansari et al. 2024). These advances make deployment evaluation more urgent: a more accurate forecast is valuable only if operations can execute the resulting plan.

2.2 Forecast-to-Decision and Predict-Then-Optimize Research

Forecast-to-decision research directly addresses why prediction accuracy and downstream decision quality can diverge. Elmachtoub and Grigas (2022) show predict-then-optimize workflows improve by training toward decision loss rather than prediction loss alone; Mandi et al. (2024) survey decision-focused learning, integrating machine learning and constrained optimization to improve decision quality under uncertainty; and Jia et al. (2025) propose scenario predict-then-optimize for online inventory routing, showing routing and replenishment decisions need more than point-forecast accuracy. This paper shares that motivation but a different goal: rather than new forecasting or end-to-end learning algorithms, it proposes an interpretable evaluation layer that lets operations teams compare existing candidate forecasts by operational consequence, not accuracy metrics alone.

2.3 Operations Management and Planning-System Architecture

Advanced planning and ERP systems architecturally separate forecasting, planning, and execution layers (Stadtler et al. 2015), and model deployment requires validation, approval, and integration cycles that take weeks or months—so when forecasting models change faster than implementation can absorb, operations teams face a governance problem, not just a technical one. Inventory theory provides the operational foundation: Silver et al. (1998) and Axsater (2015) frame inventory planning as a cost-and-service-level problem rather than a pure prediction problem, Hyndman and Koehler (2006) show forecast-error measures have known limitations, and forecast combination can reduce model-specific swings (Bates and Granger 1969). Forrester (1961) shows that industrial systems amplify instability when material and information flows respond with delays—the bullwhip effect—and the same logic applies here: coordination mechanisms (supplier partnerships, EDI, collaborative planning) reduce the planning-execution gap but do not eliminate it, because governance processes are slow to change compared to forecasting innovation. Rolling-horizon decision theory adds the final piece: Sethi and Sorger (1991) show a locally attractive action can be undesirable once future transitions are considered, which motivates this paper's greedy-to-DP comparison as a measure of planning myopia.

2.4 Research Gap and This Paper's Position

Existing research improves forecasts, combines forecasts, or optimizes downstream decisions in isolation. Fewer studies provide a practical, interpretable decision layer that compares forecasting-driven strategies by their operational consequences—inventory exposure, plan stability, execution violations, governance burden—directly, at the forecast-to-plan boundary where execution actually happens. This paper addresses that gap; the goal is not to replace forecast accuracy as a criterion but to position it as one signal among several operational objectives, evaluated where planning decisions are executed rather than where they are generated.

3 Methods

3.1 The Planning-Execution Gap: Problem and Design Rationale

Forecasting teams optimize forecast accuracy (WAPE); operations teams execute plans. These objectives often diverge: when an improved forecast requires abrupt plan changes that exceed operational capacity—supplier lead times, warehouse labor constraints, system change cycles—the forecast becomes operationally unusable despite its accuracy gains. The paper's solution is to convert all candidate forecasts into a common planning-signal form, apply the same operational metrics to every candidate, and rank strategies by those metrics rather than by WAPE. This ensures fair comparison across strategy families (accuracy-first, ensemble, smoothing, greedy, DP, budgeted-DP) and reveals when accuracy-first selection diverges from operationally-optimal selection.

Figure 1 shows this as a deployment gate. Unlike a standard forecasting pipeline (generate forecast $\rightarrow$ report WAPE $\rightarrow$ approve), the gate adds an operational evaluation step: convert forecast to plan, measure feasibility, then approve only if feasible. This figure is *how* the paper measures feasibility; Figure 3 in Section 5.5 is the complementary *decision* protocol built on those measurements.

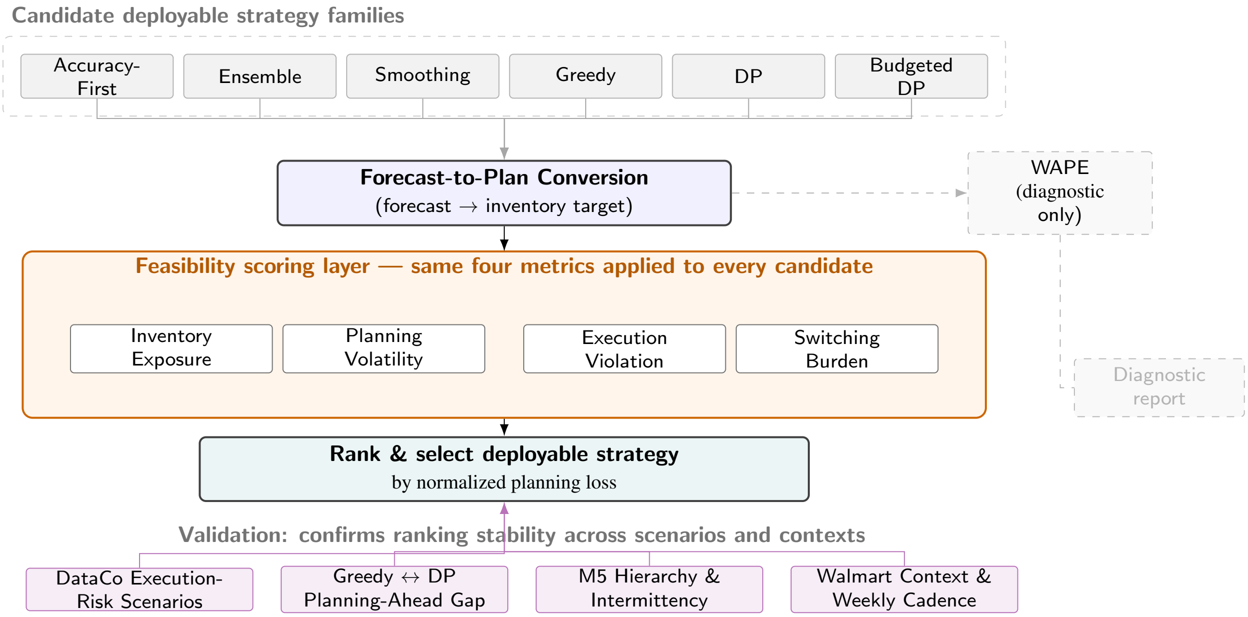

Figure 1. Forecast-to-plan feasibility gate: six strategy families (top) share one feasibility-scoring layer (orange); WAPE is tracked separately as a diagnostic only (right).

3.2 Forecast-to-Plan Conversion: Why and How

The first step converts forecasts into operational plans. Let i index a planning unit, t a planning period, and m a candidate model, so the candidate forecast is $\hat{y}^m_{i,t}$; the conversion is

$$p^m_{i,t} = \hat{y}^m_{i,t} + SS_i, \tag{1}$$

where SS_i is safety stock from demand volatility and service-level targets, and $p^m_{i,t}$ is the resulting inventory target—the operational plan.

Different forecasts create different plans: Forecast A might suggest holding 100 units and Forecast B 150, and even if both have similar WAPE, Forecast B requires 50 additional units of inventory and a larger supply-chain adjustment. Inventory targets are the right common form because they are the primary planning signal in demand-driven supply chains, cascading to procurement orders, warehouse allocations, staffing, and transportation—so measuring changes in the target captures operational workload across all of these downstream functions at once.

3.3 Feasibility Scoring: Four Operational Metrics

Once all candidates are converted to planning signals, we measure the operational consequence of each. We use four metrics:

1. Inventory Exposure (Equation 2)

$$IC^m_{i,t} = h[p^m_{i,t} - y_{i,t}]^+ + b[y_{i,t} - p^m_{i,t}]^+ \tag{2}$$

Supply chains balance two costs: holding cost h (overstocking, when $p^m_{i,t} > y_{i,t}$) and shortage cost b (understocking, when $p^m_{i,t} < y_{i,t}$), revealing whether accuracy improvements hide inventory cost increases. Realized demand $y_{i,t}$ enters only as this ex-post scoring outcome: $p^m_{i,t}$ is generated from the forecast alone (Equation 1) before $y_{i,t}$ is observed, and deployable strategies are selected on validation-period costs, never on test-period outcomes—the sole exception, the Realized-Inventory Oracle DP (Section 3.6), is non-deployable for exactly this reason.

2. Planning Volatility (Equation 3)

$$PV^m_{i,t} = |p^m_{i,t} - p_{i,t-1}| \tag{3}$$

Operations cannot replan infinitely often: suppliers need stable orders, warehouses need stable staffing, and systems need stable configurations, so sharp plan changes turn these functions into coordination bottlenecks. We measure plan movement to quantify this burden.

3. Execution Violation (Equation 4)

$$EV_{i,t}^{m} = [|p_{i,t}^{m} - p_{i,t-1}| - C_{i,t}]^{+} \tag{4}$$

Operations have a maximum absorption rate $C_{i,t}$ per period, set by supplier lead times, labor rescheduling capacity, and system update cycles. When a plan change exceeds $C_{i,t}$, the operation cannot execute it; the excess $EV_{i,t}^{m}$ is the execution violation, and this is what "planning-execution gap" means operationally.

4. Model Switching Burden (implicit in loss function): Each model change requires validation, approval, and monitoring—governance processes that take time and resources regardless of whether the new model is more accurate; we measure the number of model changes to quantify this burden. Aggregating execution violations over time gives the **planning-execution gap rate**, the fraction of periods where the operation cannot execute:

$$PEG(s) = \frac{1}{T}\sum_{t=1}^{T} \mathbb{I}\{EV_t(s) > 0\}. \tag{5}$$

A strategy with PEG rate 30% requires emergency handling, expediting, or replanning in 30% of periods; PEG 5% is operationally far more feasible.

3.3.1 Why Exclude WAPE from the Objective?

We report WAPE separately as a diagnostic metric rather than folding it into the operational objective: it measures precision, not consequence, so a 10% WAPE improvement that doubles execution violations is operationally worse despite looking better on paper, and it can mask trade-offs since a more accurate forecast often requires more frequent model changes that create governance burden invisible to accuracy metrics. Separating WAPE (diagnostic) from operational metrics (primary) lets planners see when an accuracy improvement is operationally worthless or even harmful.

3.4 Normalized Planning Loss: Combining Operational Metrics

We combine the four operational metrics into a single planning-loss objective:

$$\begin{aligned} L(s) = {} & \lambda_{\text{inventory}} \cdot \text{NIC}(s) + \lambda_{\text{volatility}} \cdot \text{NVC}(s) \\ & + \lambda_{\text{execution}} \cdot \text{NEC}(s) + \lambda_{\text{switch}} \cdot \text{NSC}(s), \end{aligned} \tag{6}$$

where NIC, NVC, NEC, NSC are normalized inventory, volatility, execution, and switching components, and λ values are organizational weights.

Normalization: All components are divided by the accuracy-first baseline (Global Best), so a normalized loss of 0.9 means 10% better than accuracy-first and 1.1 means 10% worse—meaningful even when absolute cost values differ across datasets.

Weight selection: Current weights are $\lambda_{\text{inventory}} = 1.0, \lambda_{\text{volatility}} = 0.10, \lambda_{\text{execution}} = 0.10, \lambda_{\text{switch}} = 0.05$, reflecting a supply chain where inventory cost dominates but feasibility and governance still matter; organizations can adjust these by constraint, e.g., raising $\lambda_{\text{execution}}$ under tight execution capacity or λ_{switch} under strict governance.

3.5 From Metrics to a Selection Procedure

Figure 1 assembles the equations above into one procedure. Candidate strategies first generate forecasts and convert them into planning signals (Equation 1); every candidate then passes through the same feasibility-scoring layer, where inventory exposure, planning volatility, execution violation, and switching burden are computed identically regardless of which strategy produced the candidate (Equations 2–5). These four scores are aggregated into the normalized planning loss $L(s)$ (Equation 6), which ranks strategies for deployment. A final validation pass—against execution-risk scenarios, oracle benchmarks, and the cross-dataset checks in Section 5.6—confirms that a ranking is not an artifact of one scenario or dataset before it is trusted.

The procedure's only genuine innovation is the middle step. Candidate generation and final ranking are unremarkable;

what changes is that accuracy-first, ensemble, smoothing, greedy, DP, and budgeted-DP candidates are all routed through the same scoring layer before ranking, so their normalized planning losses are directly comparable rather than each strategy family being judged by whatever metric its own literature happens to favor. The gate is agnostic to strategy family; we evaluate six. **Accuracy-First** chooses the model with lowest validation WAPE (the conventional benchmark). **Ensemble** combines candidate forecasts by simple averaging or operational-loss weighting, dampening model-specific swings while retaining both models' information. **Smoothing** blends the new plan with the previous period's,

$$p_t = \alpha p_t^{\text{candidate}} + (1-\alpha)p_{t-1}, \tag{7}$$

where lower α slows adaptation and reduces execution burden, trading stability for responsiveness. **Greedy** selection minimizes only the current-period cost $c_t(m_{t-1}, m)$ of switching from model m_{t-1} to m,

$$m_t^G = \arg\min_{m \in \mathscr{M}} c_t(m_{t-1}, m), \tag{8}$$

ignoring future consequences. **Dynamic Programming** corrects this by accounting for future transitions,

$$V_t(m_{t-1}) = \min_{m \in \mathscr{M}} \{c_t(m_{t-1}, m) + V_{t+1}(m)\}, \quad V_{T+1}(\cdot) = 0; \tag{9}$$

the greedy-to-DP gap measures the cost of this myopia, small when switching cost is low and material when approval or coordination costs are high. **Budgeted DP** adds a hard governance constraint on total switches,

$$\sum_{t=2}^{T} \mathbb{I}\{m_t \neq m_{t-1}\} \leq K, \tag{10}$$

relevant when an organization imposes a fixed annual limit on model changes.

3.6 Validation Benchmarks: Non-Deployable Oracles

Two oracle benchmarks provide diagnostic references, neither deployable. The **Realized-Inventory Oracle DP** uses realized test-period inventory outcomes over the same forecast candidates—information no real planner has at decision time—and the DP-oracle gap (its normalized-loss distance from the best deployable strategy) shows how much planning value remains unavailable without that hindsight. The **Perfect-Demand Oracle** uses realized demand itself as the planning signal, yet is often not a lower bound: exact demand tracking can require plan changes faster than execution capacity allows, producing high execution violations. We call this the *curse of accuracy*—a forecast that tracks demand perfectly can still yield an infeasible plan—and its reference gap is therefore diagnostic only, combining forecast uncertainty and feasibility effects to show that accuracy alone is insufficient.

4 Data Collection and Experimental Design

The empirical design uses DataCo, Favorita, M5, and Walmart for distinct operational roles, summarized in Table 1. DataCo is not an inventory-control dataset; it transfers empirical heterogeneity in execution risk into scenario design. Favorita is the main forecast-to-plan proof of concept, M5 tests hierarchy and intermittent-demand robustness, and Walmart provides secondary business-context robustness.

Table 1. Dataset roles and experimental purpose.

Dataset	**Role**	**Data used**	**Purpose**
DataCo	Execution-risk anchor	Late-delivery outcomes by context	Defines execution-risk scenarios; not an inventory-control dataset.
Favorita	Main proof of concept	Daily store-family demand panel	Tests forecast-to-plan conversion and strategy ranking under execution risk.
M5	Robustness analysis	Hierarchical, intermittent demand	Tests whether the gap persists across grain and demand sparsity.
Walmart	Business-context robustness	Weekly sales with holiday/markdown context	Tests context-aware forecasting, cadence, and switch-budget sensitivity.

DataCo execution-risk anchor. Supply-chain fulfillment records with late-delivery outcomes. Late-delivery anchors from context-level percentiles range from 0.535 (low-risk) to 0.901 (severe-risk); under the additive clipped scenario mapping, these produce execution weights from 0.318 to 0.500—sensitivity weights, not exact dollar costs.

Favorita proof of concept. A daily retail demand-planning panel from Corporacion Favorita treating each store-family pair as a planning unit, with promotion, holiday, oil-price, transaction, and store-context information. Candidate forecasts include transparent baselines and, when available, LightGBM and XGBoost (Ke et al. 2017; Chen and Guestrin 2016).

M5 robustness design. M5 contains 3,049 items across 10 stores (Makridakis et al. 2022); the analysis samples item-store series and evaluates item-store, department-store, and category-store planning grains and intermittent-demand buckets, testing whether the planning-execution gap persists when demand is sparse or aggregated.

Walmart business-context robustness design. Weekly store-department sales with holiday, markdown, store, and economic context fields, used to compare history-only against history-plus-context forecasts under holiday/markdown stress windows, weekly cadence constraints, and switch-budget sensitivity—testing whether the feasibility layer stays informative when planning is weekly and context-driven rather than daily. Additional local-store retail datasets are left for future robustness work.

5 Results and Discussion

5.1 Core Finding: The Planning-Execution Gap is Measurable and Structural

The analysis begins with a fundamental question: when accurate forecasts and operational feasibility diverge, how often does this matter? The answer is striking: in 91.1% of evaluation groups, the strategy that minimizes WAPE is NOT the strategy that minimizes operational planning loss, and this is structural across datasets, run modes, and execution-risk scenarios, not a marginal effect. Accuracy-first strategies have a PEG rate of 15.4% of periods (execution violations exceeding operational capacity), compared with 2.7% for the best deployable operational-loss strategy—a 12.7-percentage-point gap. For a typical 365-period planning year, that difference means accuracy-first strategies require emergency execution handling roughly 56 times versus 10 times for operationally optimized strategies: the planning-execution gap is measurable, quantifiable, and operationally consequential.

When accuracy-best and deployment-best strategies diverge, no single method is universally optimal; different strategy families serve different operational constraints, starting with ensemble methods.

5.2 Finding 1: Ensemble Methods Provide Low-Disruption First Rollout

Ensemble averaging combines candidate forecasts into one blended prediction—if Model A predicts 100 units and Model B predicts 120, the ensemble predicts 110—which dampens abrupt plan movement while retaining both models' information. In the Favorita evidence, Global Best (accuracy-first) reaches WAPE 17.7% with execution violation rate 42.1% (normalized loss 1.550), while Simple Ensemble reaches WAPE 17.4% with violation rate 5.2% (normalized loss 1.422): a 36.9pp violation reduction at no WAPE cost. Ensemble is therefore a practical first-rollout candidate wherever plan movement is costly and requires no new model validation or monitoring infrastructure, though it offers little advantage where WAPE accuracy is itself the competitive edge (e.g., predictable, high-value, low-volume demand).

Figure 2 places all nine baseline-scenario strategies on the same accuracy-execution plane. WAPE and execution penalty do not move together: Global Best and Family Best sit almost on top of each other in the upper-middle region, while Simple Ensemble and the smoothing-based strategies reach comparable or better WAPE at a fraction of the execution penalty. The Realized-Demand Oracle—the non-deployable reference that tracks realized demand exactly—has zero forecast error but the single highest execution penalty in the panel: the "curse of accuracy" introduced in Section 3, where perfect demand tracking moves the plan faster than operations can absorb.

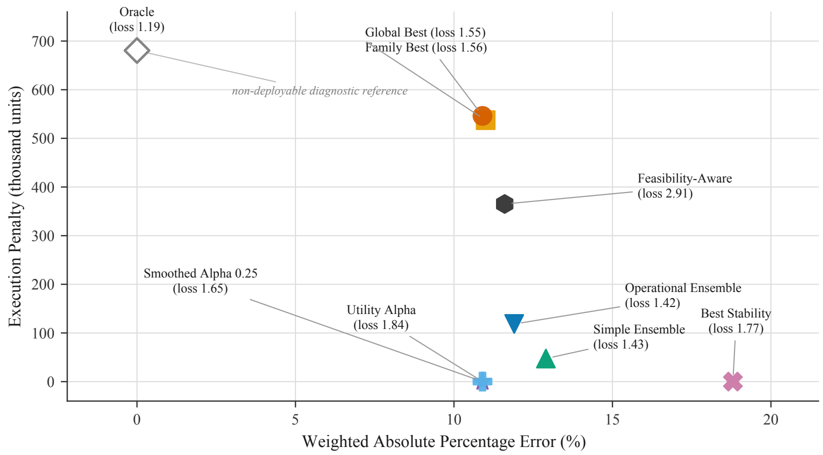

Figure 2. Favorita accuracy-execution frontier, baseline scenario: WAPE versus execution penalty for all nine deployable strategies plus the non-deployable Oracle reference, each labeled with its normalized planning loss.

5.3 Finding 2: Smoothing and DP Serve Different Governance Regimes

When ensemble's improvement is insufficient, smoothing and dynamic programming cover the remaining operational constraints. **Smoothing** slows adaptation by blending the new plan with the previous period's, trading response speed for stability: Best Stability ($\alpha \approx 0.25$) reaches WAPE 20.6% with execution violation rate 0.03% (normalized loss 1.220)—a 2.9pp WAPE cost for 99.97% execution stability. It is worth this cost only where execution capacity is so constrained that plan stability outweighs accuracy, e.g., single-source suppliers or systems already at maximum utilization.

Dynamic programming accounts for future switching costs that one-period greedy selection ignores: staying with an already-approved model can beat switching to a marginally better one once tomorrow's approval cost is counted. The greedy-to-DP gap (0.044, small when switching cost is low) and DP-oracle gap (0.198) quantify this myopia and the feasibility remaining after DP selection; DP is worth its complexity when changes require multi-step approval or system coordination, and budgeted DP adds a hard annual switching limit under strict change-control governance.

5.4 Finding 3: Planning Hierarchy Changes Optimal Strategy

The most striking finding is that different aggregation levels favor fundamentally different strategies, challenging the "one global forecasting model" paradigm. In M5, the finest grain (item-store, sparse and volatile) favors Smoothed Alpha 0.25 (WAPE 25.4%, violation 0.0%, loss 1.026) because smoothing dampens noise without model changes; aggregating to department-store demand shifts the advantage to Operational Ensemble (WAPE 15.9%, violation 12.9%, loss 0.991), and category-store aggregation keeps it (WAPE 14.9%, violation 9.6%, loss 1.033), since ensembles can now capture different models' information once demand is stable. Supply chains should therefore not deploy one global model across all planning grains: item-level favors smoothing, department- and category-level favor ensemble, with DP added where governance matters—planning-system architecture should be hierarchy-aware for exactly this reason.

5.5 Model-Rollout Protocol: From Evidence to Practice

Findings 1–3 above are diagnostic: they explain *why* strategies diverge. This section turns that diagnosis into the paper's primary practical deliverable—a five-step protocol that a planning team can follow to go from a set of candidate forecasts to a specific, defensible deployment decision.

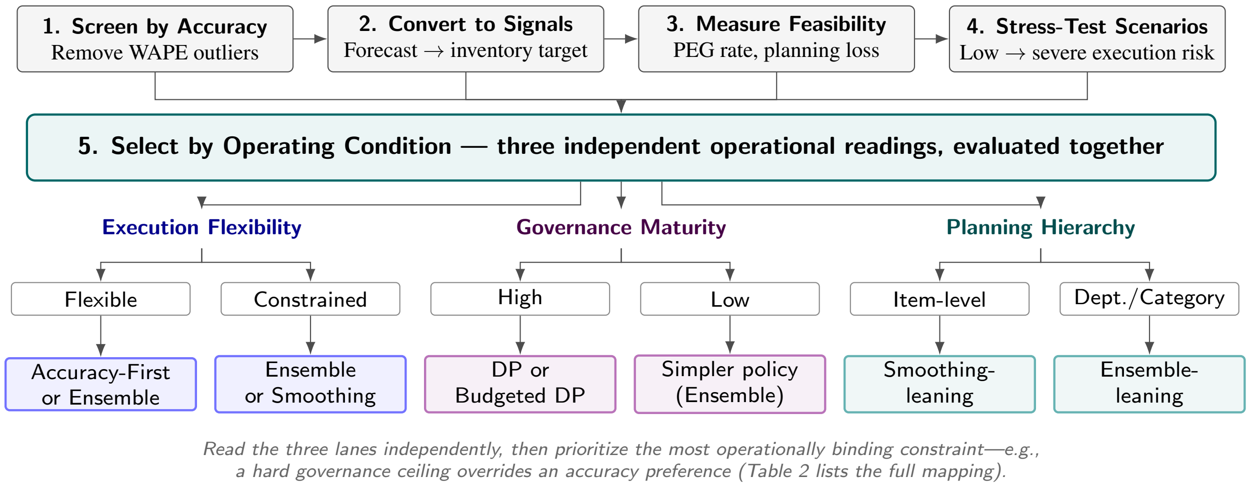

Figure 3. Model-rollout protocol, the paper's primary practical output: Steps 1–4 screen and score every candidate, and Step 5 combines three operational lanes—execution flexibility, governance maturity, planning hierarchy—into a strategy recommendation.

Steps 1–4, shown across the top of Figure 3, are a short mechanical pipeline: screen out clearly weak forecasts (WAPE $> 30\%$), convert each survivor into an inventory target using organization-specific policies, measure its PEG rate and normalized planning loss, and stress-test it across execution-risk scenarios from low to severe. Step 5 is the decision core, combining three operational readings that were each established by a Finding above: execution flexibility (Finding 1) points toward accuracy-first or ensemble when flexible, ensemble or smoothing when constrained; governance maturity (Finding 2) points toward DP or budgeted DP when switching approval is costly, simpler policies otherwise; and planning hierarchy (Finding 3) points toward smoothing at fine grain, ensemble at coarser grain. Table 2 gives the complete condition-to-policy mapping behind each leaf.

Table 2. Deployment strategy families and rollout conditions.

Operating condition	**Policy**	**Reason**
Flexible execution capacity	Accuracy-first or ensemble	WAPE gains are useful when plan movement is absorbable, but feasibility screening is still required.
Moderate execution constraint	Ensemble	Averaging dampens model-specific swings while retaining adaptation.
Severe execution constraint	Smoothing	Gradual adaptation reduces execution burden when abrupt movement is costly.
High switching governance	DP or budgeted DP	Horizon-aware selection accounts for future switching and coordination burden.
Multi-level hierarchy	Choose by planning grain	Item, department, and category planning favor different mechanisms.

The protocol is structured evaluation, not single-method prescription: it provides a repeatable procedure, not a universal answer, and the final decision still depends on organizational constraints and risk tolerance. Section 5.6 tests whether this procedure holds up outside the conditions it was built on.

5.6 Robustness and Validation

Four complementary checks confirm that findings are structural, not data artifacts, and that the protocol in Section 5.5 is trustworthy outside the conditions it was built on.

5.6.1 Baseline and External Validation: Is This a Real Problem?

The 91.1% mismatch rate and 12.7pp PEG-rate gap demonstrate that WAPE-best selection and deployment-best selection answer fundamentally different questions; the study does not claim feasibility-aware strategies always dominate, only that the operational deployment criterion is measurably distinct from the accuracy criterion. This divergence is not unique to the present data—it replicates a pattern already established in the predict-then-optimize literature, where Elmachtoub and Grigas (2022) show that prediction-loss-minimizing models are not necessarily loss-minimizing for the downstream decision, and Jia et al. (2025) report the same divergence for inventory-routing decisions. This paper's contribution is not the divergence itself but a feasibility-specific metric (PEG rate) and an evaluation layer that makes it measurable for demand planning specifically, where prior work has largely addressed routing and pricing decisions instead.

5.6.2 Scenario Validation: Does Execution Pressure Change Rankings?

DataCo-informed execution-risk scenarios validate whether tighter execution constraints change strategy rankings: as execution-risk weights increase from low to severe, ensemble and smoothing methods become relatively more attractive than accuracy-first selection, supporting scenario testing as a deployment practice since the optimal model for a flexible environment may not be optimal for a constrained one.

Figure 4 orders the five scenarios by increasing severity and tracks all nine strategies through each; the y-axis is broken because Feasibility-Aware, an early naive selector, sits nearly twice as high as everyone else and would otherwise compress the eight competitive strategies into an unreadable band. Among those eight, Global Best and Family Best rise fastest as execution risk tightens (1.55 at baseline to 2.00 at severe), while Simple Ensemble, Operational Ensemble, and the smoothing-based strategies stay comparatively flat—direct visual evidence that accuracy-first selection is the strategy most exposed to execution-risk escalation.

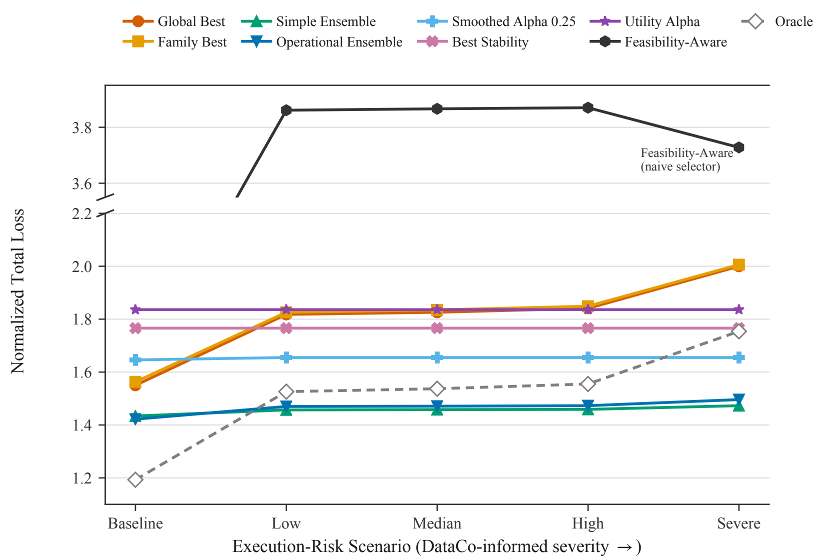

Figure 4. Favorita normalized planning loss across execution-risk scenarios, ordered by increasing severity. The broken y-axis separates Feasibility-Aware (top panel) from the eight competitive strategies (bottom panel).

Algorithmic validation asks the complementary question of whether planning ahead has value: the greedy-to-DP gap (0.044) measures the cost of myopic one-period selection, small when simple policies are adequate and material when switching governance costs (approval workflows, system change cycles) make DP's planning-ahead value important—confirming DP is a deliberate choice for high-governance environments, not a universal improvement.

5.6.3 Cross-Dataset Robustness Validation

Favorita (daily granularity, rich context) is the main proof of concept; M5 (hierarchical, sparse demand) is re-evaluated under the same DataCo-calibrated severity scenarios as Favorita and extends the finding by showing that hierarchy and sparsity genuinely change the optimal strategy rather than adding noise; Walmart (weekly cadence, business context) confirms the same 80.0% accuracy-first mismatch rate at a different temporal frequency, showing the gap is not limited to daily retail grocery planning. Together, the four checks support the thesis that feasibility-constrained planning is a multi-objective tradeoff surface, not a single-method leaderboard, and that each lane of the Section 5.5 protocol rests on evidence rather than assumption: baseline validation establishes that the gap itself is real, scenario and algorithmic validation justify the execution-flexibility and governance-maturity lanes specifically, and cross-dataset validation justifies the planning-hierarchy lane. A planner who follows Figure 3 is relying on a procedure stress-tested against the operational conditions it is meant to handle, not on advice tuned to a single dataset.

5.7 Limitations and Future Work

Public datasets do not observe the full internal cost of execution changes, planner overrides, or system reconfiguration, so execution capacity is proxied rather than directly measured; internal operational data could improve this calibration. The study also focuses on model-selection strategy rather than real-time plan adjustment or human planner intervention—extending the framework to planner workflows and collaborative replanning is a natural next step.

6 Conclusion

This paper reframes operational demand planning as feasibility-constrained model selection through a forecast-to-plan evaluation gate. The central finding is that in 91.1% of evaluation groups, the accuracy-best deployable strategy diverges from the operational-loss-optimal deployable strategy. The accuracy-first execution violation rate (15.4% of periods) is 5.7 times higher than the best deployable operational-loss strategy (2.7% of periods).

Section 1.1 opened with four operational symptoms, and each now has a direct counterpart above rather than a generic claim of improvement. *Execution violations* are what the PEG-rate metric catches (15.4% versus 2.7%). *Governance burden* is addressed by the protocol's governance-maturity lane, which recommends DP or budgeted DP specifically where switching requires approval, on the empirical basis of the greedy-to-DP gap. *Supplier disruption* falls where planning volatility falls—Simple Ensemble cuts execution violations by roughly 37 percentage points at no WAPE cost, though other cases require a small WAPE cost for stability, which the execution-flexibility lane makes explicit. *Operational gridlock* is addressed structurally: the five-step protocol replaces ad hoc negotiation with a repeatable, auditable procedure. Each symptom now has a metric and a documented response, which is what was missing before.

Implementation Agenda

Adopting a feasibility-aware planning system requires the three organizational capabilities in Table 3. Success should be tracked across all three metric families together—forecasting (WAPE, MAE), planning (PEG rate, normalized loss), and operational (supplier rework frequency, planner coordination time)—so that feasibility gains are visible to the teams who have to approve them, not only to the team that built the model.

Table 3. Implementation agenda for a feasibility-aware planning system.

Capability	Key actions
Data & measurement	Estimate execution capacity from lead times and change cycles; log exception periods; calibrate violation cost.
Governance design	Define approval thresholds for WAPE, violation rate, and planning loss; stress-test before roll-out; align forecasting and operations on shared criteria.
System integration	Convert forecast-to-plan at the system boundary; embed capacity constraints in optimization; monitor violations and loss alongside WAPE.

This work bridges forecasting advancement and operational execution: forecast accuracy is necessary but not sufficient for deployment. Making execution feasibility measurable and comparable across strategies lets organizations balance accuracy, stability, and operational realism, and adopt incrementally, starting with ensemble methods and progressing to DP or budgeted DP as governance maturity increases. The goal is not to replace forecasting metrics but to evaluate them where planning decisions are actually executed.

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Biography

Yihua Xu is an independent researcher and AI and Data professional based in San Jose, California, USA. She currently works at Accenture in the AI and Data practice, with professional interests spanning machine learning, operations research, supply chain analytics, and decision intelligence. She received her Bachelor of Science degree in Industrial Engineering from the H. Milton Stewart School of Industrial and Systems Engineering at the Georgia Institute of Technology and her Master of Business Analytics degree from the MIT Sloan School of Management at the Massachusetts Institute of Technology. Her research focuses on feasibility-aware decision frameworks for operational planning systems, including demand planning, forecast-to-plan conversion, planning stability, execution-risk evaluation, and reverse logistics. Her work aims to connect predictive models with executable operational decisions under inventory, capacity, and planning-governance constraints.

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